STPM 2016 PHYSICS 960/3

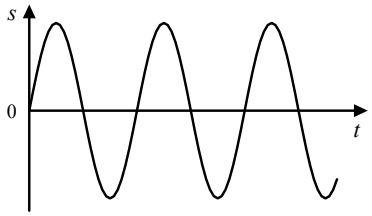
Questions and Answers

Values of constants: $h = 6.63 \times 10^{-34}$ *J s,* $c = 3.00 \times 10^{8}$ *m* s^{-1}, $e = 1.60 \times 10^{-19}$ *C,* $1\ u = 1.66 \times 10^{-27}$ *kg* $\equiv 931.5$ *MeV,* $N_A = 6.02 \times 10^{23}$ mol^{-1}.

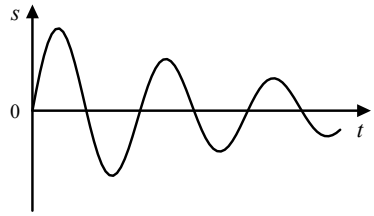
Section A [15 marks]

1. An oscillating spring-mass system is underdamped. Which graph shows the variation of the displacement, *s*, with time, *t*, for the motion?

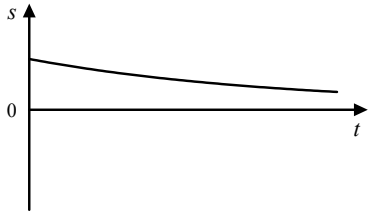
A



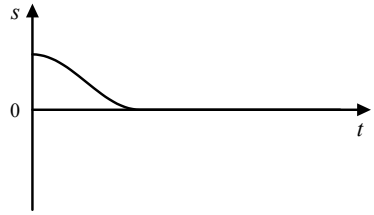
B



C



D



Answer: B

An underdamped system oscillates with an amplitude that decreases gradually (exponentially) with time: graph B.

Explanation: A is undamped, C is overdamped (slow return without oscillation) and D is critically damped.

2. A pendulum with a natural frequency, f_1, is driven by a periodic driving force of frequency, f_2. Which combination is true when resonance occurs?

	Amplitude	*Frequency*
A	Maximum	$f_2 = f_1$
B	Minimum	$f_2 = f_1$
C	Maximum	$f_2 = 2f_1$
D	Minimum	$f_2 = 2f_1$

Answer: A

At resonance the driving frequency equals the natural frequency ($f_2 = f_1$) and the amplitude is maximum.

Explanation: Energy is transferred most efficiently when the driver is in step with the natural oscillation.

3. A progressive wave on a string is represented by the equation $y = 0.25 \sin (500t - 15x)$, where x and y are in metres and t is in seconds. What is the speed of a particle of the string at $x = 0.10$ m when $t = 4.0$ s?

A 33 m s^{-1} B 54 m s^{-1} C 110 m s^{-1} D 120 m s^{-1}

Answer: C

$$v = \frac{\partial y}{\partial t} = (0.25)(500) \cos (500t - 15x) = 125 \cos (2000 - 1.5)$$

$$\cos (1998.5 \text{ rad}) = 0.90 \Longrightarrow v = 113 \text{ m s}^{-1} \approx 110 \text{ m s}^{-1}$$

Explanation: The particle velocity is the time derivative of the displacement, not the wave speed ($500/15 = 33 \text{ m s}^{-1}$, option A). Use radians.

4. Two similar progressive waves travelling in the opposite directions with wavelengths of 1.0 m, frequencies of 5.0 Hz and amplitudes of 0.20 m respectively. At time $t = 0$, the displacement at x is zero. What is the expression of a resultant wave when the two progressive waves superposed?

A $0.2 \cos 2\pi x \sin 10\pi t$ B $0.2 \cos 10\pi x \sin 2\pi t$

C $0.4 \cos 2\pi x \sin 10\pi t$ D $0.4 \cos 10\pi x \sin 2\pi t$

Answer: C

$$k = \frac{2\pi}{1.0} = 2\pi, \ \omega = 2\pi(5.0) = 10\pi$$

$$y = 2(0.20) \cos 2\pi x \sin 10\pi t = 0.4 \cos 2\pi x \sin 10\pi t$$

Explanation: Two identical waves in opposite directions give a standing wave of amplitude $2A \cos kx$.

5. Which statement is true about electromagnetic and sound waves?

A The electromagnetic wave can be polarised but the sound wave cannot be polarised.
B The sound wave shows Doppler effect but the electromagnetic wave does not.
C Both waves propagate energy through vibrations of particles.
D Both waves require a medium for propagation.

Answer: A

Electromagnetic waves are transverse and can be polarised; sound waves in air are longitudinal and cannot.

Explanation: Both show the Doppler effect; electromagnetic waves need no medium and are not vibrations of particles.

6. The intensity level of a sound at a distance of 50 cm from a point source is 65 dB. What is the power of the sound produced by the source?
[The threshold intensity of hearing is 1.0×10^{-12} W m^{-2}.]

A 1.0×10^{-6} W B 2.5×10^{-6} W C 3.2×10^{-6} W D 9.9×10^{-6} W

Answer: D

$$I = (1.0 \times 10^{-12})10^{6.5} = 3.16 \times 10^{-6} \text{ W m}^{-2}$$
$$P = 4\pi r^2 I = 4\pi(0.50)^2(3.16 \times 10^{-6}) = 9.9 \times 10^{-6} \text{ W}$$

Explanation: Convert 50 cm to 0.50 m.

7. A person who has a long-sighted vision can only see clearly when objects are at a distance of 90.0 cm and beyond. For a normal vision, objects can be seen clearly at a distance as near as 25.0 cm. What is the focal length of the corrective lens that enables the person to have a normal vision?

A −34.6 cm B −19.6 cm C +19.6 cm D +34.6 cm

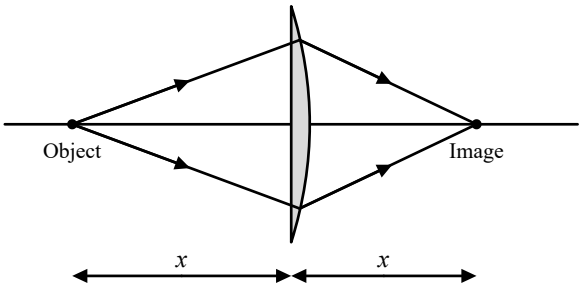
Answer: D

The lens must form a virtual image at 90.0 cm of an object at 25.0 cm:

$$\frac{1}{f} = \frac{1}{25.0} + \frac{1}{-90.0} \Longrightarrow f = +34.6 \text{ cm}$$

Explanation: Long sight is corrected with a converging (positive) lens.

8. An object is placed at a distance, x, in front of a thin plano-convex lens so that the image is formed at an equal distance, x, behind the lens is shown in the diagram below.



If the refractive index of the lens is 1.5 and the radius of curvature of the curved surface is 20 cm, what is the value of x?

A 10 cm B 20 cm C 40 cm D 80 cm

Answer: D

$$\frac{1}{f} = (1.5 - 1)\frac{1}{20} \Longrightarrow f = 40 \text{ cm}$$

Equal object and image distances occur at $u = v = 2f \Longrightarrow x = 80$ cm

Explanation: From $1/x + 1/x = 1/f$, $x = 2f$.

9. A beam of non-polarised light passes through two polarisers. The angle between the polarising axes of the polarisers is 30°. What is the percentage of the intensity of the non-polarised incident beam which is absorbed by the two polarisers when the light passes through the two polarisers?

A 25% B 38% C 63% D 75%

Answer: C

$$\text{Transmitted} = \frac{1}{2}\cos^2 30° = 0.375 \text{ of } I_0$$

$$\text{Absorbed} = 1 - 0.375 = 0.625 \approx 63\%$$

Explanation: The first polariser absorbs half the unpolarised light; Malus's law applies at the second.

10. A flashlight emits a beam of light with a power of 3.0 W and an average wavelength of 550 nm. How many photons are freed by the flashlight per second?

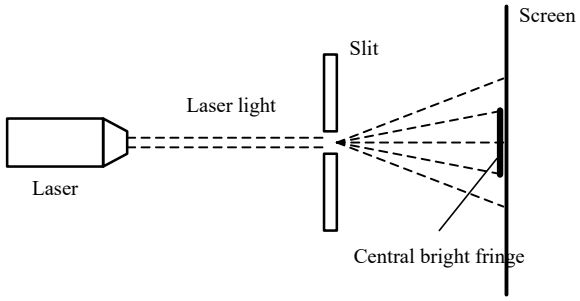
A $2.8 \times 10^{10}\ s^{-1}$ B $8.3 \times 10^{18}\ s^{-1}$ C $1.5 \times 10^{25}\ s^{-1}$ D $2.5 \times 10^{27}\ s^{-1}$

Answer: B

$$N = \frac{P\lambda}{hc} = \frac{(3.0)(550 \times 10^{-9})}{(6.63 \times 10^{-34})(3.00 \times 10^{8})} = 8.3 \times 10^{18}\ s^{-1}$$

Explanation: Number of photons per second = power ÷ energy per photon.

11. The aerial view of a single slit diffraction experiment set up using laser light of a certain wavelength is shown in the diagram below.



The width of the central bright fringe will be smaller if

A the wavelength is longer and the slit is wider
B the wavelength is shorter and the slit is wider
C the wavelength is longer and the slit is narrower
D the wavelength is shorter and the slit is narrower

Answer: B

Width of the central fringe = $\frac{2\lambda D}{a}$: smaller for a shorter wavelength and a wider slit

Explanation: Diffraction spreads light more when the slit is narrow compared with the wavelength.

12. The total energy of electrons in the nth orbit in a hydrogen atom is given by $E_n = -\frac{me^4}{8\varepsilon_0{}^2h^2n^2}$, where m and e are the mass and the charge of an electron respectively. What is the de Broglie wavelength of the electron in the nth orbit?

A $\frac{2\varepsilon_0 h^2 n}{me^2}$ B $\frac{8\varepsilon_0{}^2 h^2 cn^2}{me^4}$ C $\frac{\sqrt{2}\,\varepsilon_0 h^2 n}{me^2}$ D $\frac{\sqrt{8}\,\varepsilon_0{}^2 h^2 cn^2}{me^4}$

Answer: A

$$K = -E_n = \frac{me^4}{8\varepsilon_0{}^2h^2n^2};\ p = \sqrt{(2mK)} = \frac{me^2}{2\varepsilon_0 hn}$$

$$\lambda = \frac{h}{p} = \frac{2\varepsilon_0 h^2 n}{me^2}$$

Explanation: For a Bohr orbit the kinetic energy equals minus the total energy. (Check: this equals $2\pi r_n/n$.)

13. The characteristic lines in the X-ray spectrum is produced when

A an electron at the highest energy level is captured into the nucleus
B an electron in the target atom makes a transition to the innermost orbit
C the incident electron undergoes deceleration and radiates all its energy as a photon
D the incident electron successfully knocks out an electron from the outermost orbit of the target atom

Answer: B

Characteristic lines are emitted when an electron of the target atom drops into a vacancy in an inner shell (innermost orbit) created by the bombarding electrons.

Explanation: Deceleration of incident electrons produces the continuous spectrum, not the lines.

14. What is the mass defect of $^{12}_{6}C$ nucleus?

[Mass of $^{12}_{6}C$ atom = 12.0000 u, mass of $^{1}_{1}H$ atom = 1.0078 u, mass of neutron = 1.0087 u.]

A 0.0108 u B 0.0936 u C 0.0990 u D 0.1044 u

Answer: C

$$\Delta m = 6(1.0078) + 6(1.0087) - 12.0000 = 0.0990 \text{ u}$$

Explanation: Hydrogen-atom masses are used so the electrons balance with the carbon atom's 6 electrons.

15. The function of the control rod in a nuclear fission reactor is to

A absorb the excess neutrons so that a controlled chain reaction can occur
B produce neutrons so that a chain reaction can occur
C slow down the neutrons produced in the reaction
D accelerate the neutrons produced in the reaction

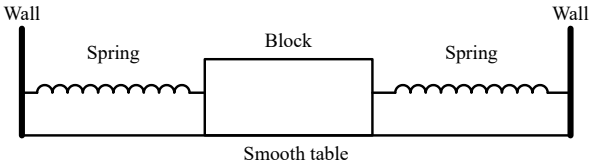
Answer: A

Control rods absorb excess neutrons so that a controlled (steady) chain reaction occurs.

Explanation: The moderator slows the neutrons; it does not absorb them.

Section B [15 marks]

16. A block on a smooth table is attached to two stretched identical springs which are fixed to the walls are shown in the diagram below.



The mass of the block is 1.5 kg. The initial extension of each spring is 5.0 cm. The effective force constant for the combined springs is 25 N m^{-1}. The block is then displaced 4.0 cm to the left and released to perform a simple harmonic motion. Calculate

(a) the period of the simple harmonic motion, [2]

Answer

$T = 2\pi\sqrt{(\frac{m}{k})} = 2\pi\sqrt{(\frac{1.5}{25})}$ 1

$T = 1.54$ s 1

2

Explanation: Use the effective force constant of the two springs.

(b) the maximum velocity of the block, [3]

Answer

$\omega = \sqrt{(\frac{k}{m})} = \sqrt{(\frac{25}{1.5})} = 4.08$ rad s^{-1} 1

$v_{max} = \omega A = (4.08)(0.040)$ 1

$v_{max} = 0.163$ m s^{-1} 1

3

Explanation: The amplitude is the initial displacement, 4.0 cm.

(c) the potential energy of the system when the block is 2.0 cm from the mid point. [3]

Answer

Each spring: $k = 25/2 = 12.5$ N m^{-1}; at 2.0 cm from the mid point the extensions are $5.0 - 2.0 = 3.0$ cm and $5.0 + 2.0 = 7.0$ cm — 1

$U = \frac{1}{2}(12.5)(0.030)^2 + \frac{1}{2}(12.5)(0.070)^2$ — 1

$U = 3.63 \times 10^{-2}$ J — 1

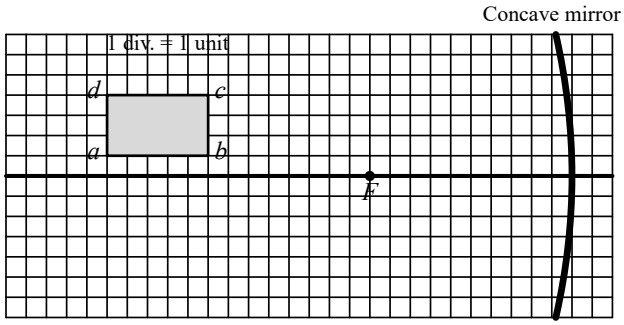
[Measured from the equilibrium position, the SHM potential energy is $\frac{1}{2}(25)(0.020)^2 = 5.0 \times 10^{-3}$ *J.]*

3

Total for question **8**

Explanation: Both springs are already stretched at equilibrium, so the system stores elastic energy even at the mid point.

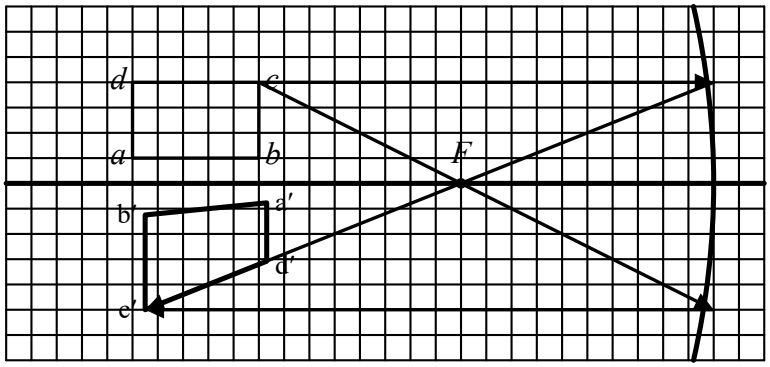
17. A rectangular object *abcd* which is placed in front of a concave mirror is shown in the diagram below.



The dimensions of the objects are 5 units × 3 units and F is the focal point of the concave mirror.

(a) Draw a ray diagram to locate the positions $a'b'c'd'$ of the image which correspond to points $abcd$ of the object on the diagram. (Sketch it on paper and attach a photo.) [2]

Answer



Correct rays from the corners (parallel ray reflected through F; ray through F reflected parallel)	1
Image $a'b'c'd'$ located below the axis between about 17.7 and 22.5 units from the mirror	1
	2

Explanation: By calculation (f = 10 units): b, c at 18 units give v = 22.5 units, m = −1.25; a, d at 23 units give v = 17.7 units, m = −0.77.

(b) State the characteristics of the image. [2]

Answer

Real and inverted	1
Distorted: the near side $b'c'$ is magnified (×1.25) while the far side $a'd'$ is diminished (×0.77); the image is reversed left to right	1
	2

Explanation: Different parts of the object are at different distances, so they have different magnifications.

(c) Estimate the ratio of $\frac{\text{area of the image}}{\text{area of the object}}$. [3]

Answer

Image: a trapezium of width 22.5 − 17.7 = 4.8 units, with parallel sides 3.75 and 2.31 units	1
Area = $\frac{1}{2}(3.75 + 2.31)(4.8)$ = 14.6 square units; object area = 15 square units	1
Ratio ≈ 1.0 (0.97)	1
	3
Total for question	**7**

Explanation: An estimate from a careful drawing on the grid should be close to 1.

Section C [30 marks]

18. (a) State **two** necessary conditions for the formation of standing waves. [2]

Answer

Two waves of the same type, frequency (wavelength), speed and amplitude	1
travelling in opposite directions along the same line superpose	1
	2

Explanation: Usually an incident wave and its reflection.

(b) A standing wave on a stretched string is represented by $y = 3.0 \cos(0.10\pi x) \sin(85\pi t)$, where y and x are in centimetres, and t is in seconds.

(i) Determine the distance between two consecutive nodes. [3]

Answer

Nodes where $\cos(0.10\pi x) = 0$: $0.10\pi x = \frac{\pi}{2}, \frac{3\pi}{2}, \ldots$	1
x = 5 cm, 15 cm, …	1
Distance between consecutive nodes = 10 cm	1
	3

Explanation: This equals $\lambda/2$, with $\lambda = 2\pi/k$ = 20 cm.

(ii) Calculate the velocity of the waves which formed the standing wave. [3]

Answer

$\omega = 85\pi$ rad s^{-1}, $k = 0.10\pi$ cm^{-1} 1

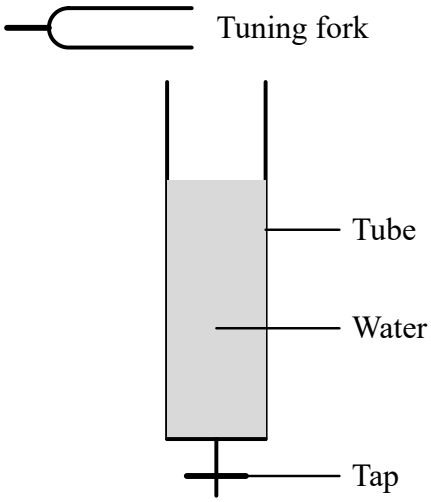
$v = \frac{\omega}{k} = \frac{85\pi}{0.10\pi}$ 1

$v = 850$ cm s^{-1} $= 8.5$ m s^{-1} 1

3

Explanation: This is the speed of each of the progressive waves forming the standing wave.

(c) A long tube which has a tap fitted to it, and filled with water is shown in the diagram below.

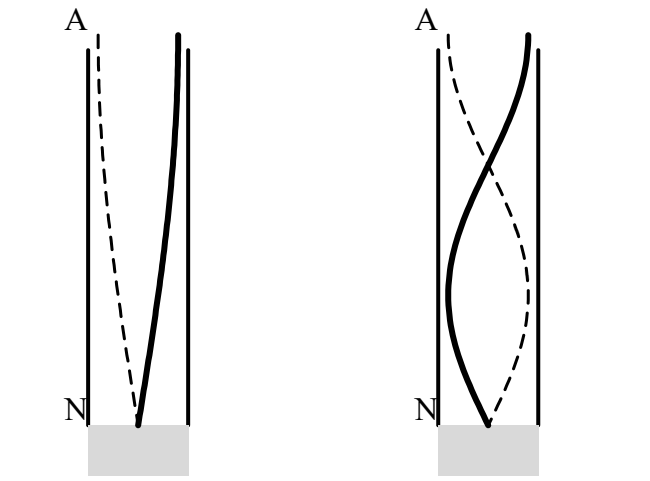


A tuning fork, of frequency 512 Hz, is sounded above the tube as the water flows out slowly through the tap. The first loud sound is heard when the length of the air column is 15.7 cm. As the

water continues to flow out, a second loud sound is heard when the length of the air column is 48.1 cm.

(i) Sketch the stationary waves in the tube when the loud sounds are heard. [2]

Answer



first resonance (λ/4) second resonance (3λ/4)

First loud sound: node at the water surface, antinode at (just above) the top: ¼λ	1
Second loud sound: ¾λ in the tube	1
	2

Explanation: The water surface is a closed end (node); the open top is an antinode.

(ii) What is meant by an *end-correction*? [1]

Answer

The antinode at the open end forms slightly outside the tube; the end-correction is the distance from the open end to this antinode	1
	1

Explanation: It is about 0.6 times the radius of the tube.

(iii) Determine the velocity of sound and the end-correction for the tube. [4]

Answer

$\frac{\lambda}{2} = 48.1 - 15.7 = 32.4$ cm $\Longrightarrow \lambda = 64.8$ cm — 1

$v = f\lambda = (512)(0.648) = 332$ m s^{-1} — 1

$\frac{\lambda}{4} = 15.7 + e \Longrightarrow 16.2 = 15.7 + e$ — 1

$e = 0.50$ cm — 1

4

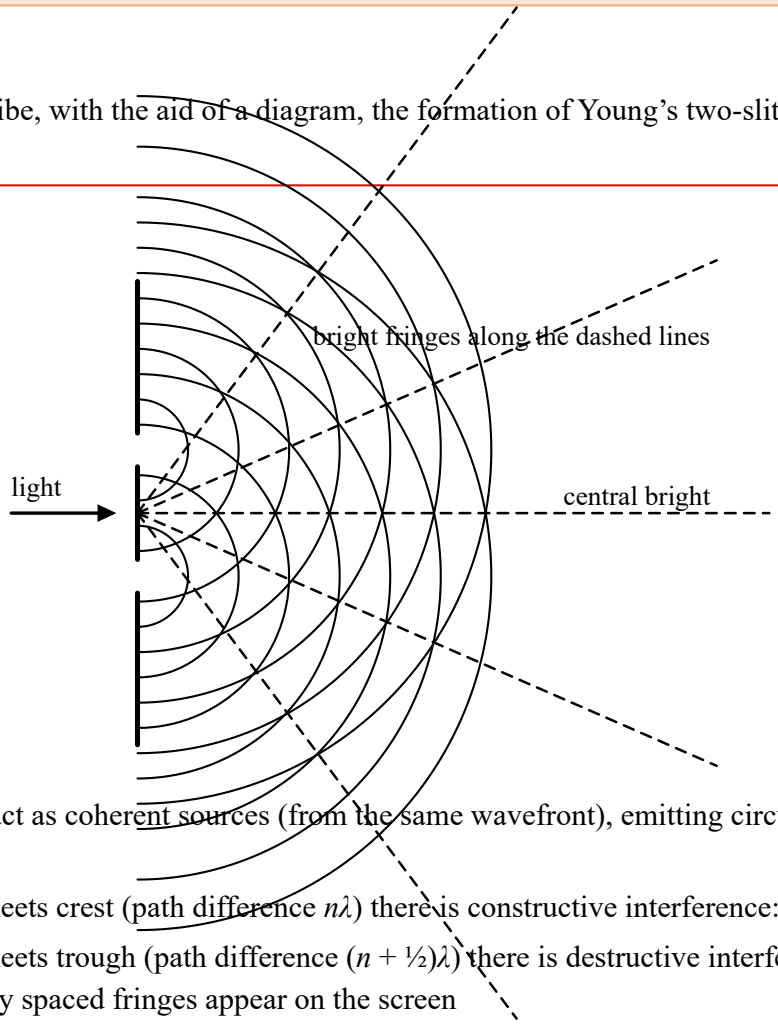
Total for question **15**

Explanation: Subtracting the two resonance lengths eliminates the end-correction.

19. (a)

(i) Describe, with the aid of a diagram, the formation of Young's two-slit interference fringes. [3]

Answer



The two slits act as coherent sources (from the same wavefront), emitting circular wavelets that overlap — 1

Where crest meets crest (path difference $n\lambda$) there is constructive interference: bright fringes — 1

Where crest meets trough (path difference $(n + ½)\lambda$) there is destructive interference: dark fringes; equally spaced fringes appear on the screen — 1

3

Explanation: The fringes are equally spaced because the path difference changes uniformly across the screen.

(ii) Derive a formula for the fringe separation, x, for a monochromatic light of wavelength, λ, if the separation of the slits is a and the distance between the slits and the screen is D. Assume

that $D \gg a$. [4]

Answer

Path difference to a point P at distance y from the centre: $\Delta = a \sin\theta$ — 1

For $D \gg a$: $\sin\theta \approx \tan\theta = \frac{y}{D} \Longrightarrow \Delta = \frac{ay}{D}$ — 1

Bright fringe: $\frac{ay_n}{D} = n\lambda \Longrightarrow y_n = \frac{n\lambda D}{a}$ — 1

Fringe separation $x = y_{n+1} - y_n = \frac{\lambda D}{a}$ — 1

4

Explanation: The small-angle approximation is valid because the screen is far away compared with the slit separation.

(iii) When the monochromatic light of wavelengths, λ_1 and λ_2, ($\lambda_1 > \lambda_2$) are used in Young's two-slit experiment, the second bright fringe for λ_1 overlaps with the third bright fringe for λ_2. Determine the ratio of λ_1 to λ_2. [2]

Answer

$\frac{2\lambda_1 D}{a} = \frac{3\lambda_2 D}{a}$ — 1

$\lambda_1 : \lambda_2 = 3 : 2$ — 1

2

Explanation: The overlapping fringes are at the same distance from the centre.

(b)

(i) A monochromatic light of wavelength, λ, is incident normally onto a transparent thin film in air. The refractive index and the thickness of the film are n and t respectively. Deduce the condition for destructive interference for the reflected light. [3]

Answer

Reflection at the top surface (air → film) has a phase change of π (half a wavelength); at the bottom surface (film → air) there is none — 1

Extra optical path in the film = $2nt$ — 1

Destructive interference: $2nt = m\lambda$, $m = 1, 2, 3, \ldots$ — 1

3

Explanation: Because of the extra half-wavelength phase change, a whole number of wavelengths gives cancellation.

(ii) Explain the function of the thin film in a non reflecting lens. [3]

Answer

A thin coating (refractive index between those of air and glass) is applied to the lens	1
Its thickness is chosen ($\lambda/4n$) so that light reflected from the top and bottom surfaces of the film is in antiphase	1
The reflected waves interfere destructively, reducing reflection and increasing the light transmitted through the lens	1
	3
Total for question	**15**

Explanation: Both reflections have the same phase change, so only the path difference ½λ matters.

20. (a) Radioactivity is a random and spontaneous disintegration of unstable nuclei. What is meant by *random* and *spontaneous* in the statement above? [2]

Answer

Random: it is impossible to predict which nucleus will decay next or when a particular nucleus will decay	1
Spontaneous: the decay is not affected by external factors such as temperature, pressure or chemical state	1
	2

Explanation: Only the probability of decay (the decay constant) is fixed.

(b) An alpha decay of Uranium-238 is represented by $^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th} + {}^{4}_{2}\text{He}$. The nucleus of the uranium which is initially at rest emits an alpha particle and produces a daughter nucleus $^{234}_{90}\text{Th}$ recoils. Calculate the

[Mass of $^{238}_{92}\text{U}$ is 238.050786 u, mass of $^{234}_{90}\text{Th}$ is 234.043583 u, mass of $^{4}_{2}\text{He}$ is 4.002603 u.]

(i) energy released in the decay, [3]

Answer

$\Delta m = 238.050786 - (234.043583 + 4.002603) = 0.004600$ u	1
$E = (0.004600)(931.5) = 4.28$ MeV	1
$E = (4.28 \times 10^{6})(1.60 \times 10^{-19}) = 6.86 \times 10^{-13}$ J	1
	3

Explanation: This energy is shared as kinetic energy between the α particle and the recoiling thorium nucleus.

(ii) velocity of the alpha particle emitted in the decay. [3]

Answer

Momentum conservation: $m_\alpha v_\alpha = m_{Th} v_{Th}$, so $K_\alpha = E\frac{m_{Th}}{m_{Th} + m_\alpha}$ 1

$K_\alpha = (6.86 \times 10^{-13})\frac{234.04}{238.05} = 6.74 \times 10^{-13}$ J 1

$v_\alpha = \sqrt{(\frac{2(6.74 \times 10^{-13})}{(4.002603)(1.66 \times 10^{-27})})} = 1.42 \times 10^7$ m s^{-1} 1

3

Explanation: The light α particle carries almost all of the kinetic energy (ignoring the recoil gives 1.44×10^7 m s^{-1}).

(c) Carbon-14 can be used for determining the age of an artefact. Assuming, a piece of wood has been recovered from an excavation, and 8.50 g of carbon is prepared. The ratio of carbon-14 to carbon-12 is 1.25×10^{-12}. The activity of the sample is found to be 42 per minute. The half-life of carbon-14 is 5760 years. Calculate the

[Given that 1 year is 3.154×10^7 s.]

(i) number of carbon-14 atoms originally present in the sample, [2]

Answer

Number of carbon atoms = $\frac{8.50}{12} \times 6.02 \times 10^{23} = 4.26 \times 10^{23}$ 1

$N_0 = (1.25 \times 10^{-12})(4.26 \times 10^{23}) = 5.33 \times 10^{11}$ 1

2

Explanation: The ratio C-14 : C-12 in living wood gives the original number of C-14 atoms.

(ii) number of carbon-14 atoms in the sample, [3]

Answer

$\lambda = \frac{\ln 2}{(5760)(3.154 \times 10^7)} = 3.82 \times 10^{-12}$ s^{-1} 1

$A = \frac{42}{60} = 0.70$ s^{-1} 1

$N = \frac{A}{\lambda} = 1.83 \times 10^{11}$ 1

3

Explanation: Convert the activity to decays per second and the half-life to seconds.

(iii) age of the wood. [2]

Answer

$N = N_0 e^{-\lambda t} \Longrightarrow t = \dfrac{\ln (N_0/N)}{\lambda} = \dfrac{\ln (5.33/1.83)}{3.82 \times 10^{-12}}$ 1

$t = 2.80 \times 10^{11}$ s $\approx 8.86 \times 10^{3}$ years 1

2

Total for question **15**

Explanation: The wood is about 1.5 half-lives old.